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#### 4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

03:10

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked\_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

#### Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked  
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC  
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina  
Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

**Note:** Refer to the visible test case for better reference.

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# 1. Calculate the survival rate by class
print(data.groupby('Pclass')['Survived'].mean())

# 2. Calculate the survival rate by embarked location
print(data.groupby('Embarked_S')['Survived'].mean())

# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize')['Survived'].mean())

# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())

# 5. Get the average fare by class
print(data.groupby('Pclass')['Fare'].mean())

# 6. Get the average age by class
print(data.groupby('Pclass')['Age'].mean())

# 7. Get the average age by survival status
```

```
print(data.groupby('Survived')['Age'].mean())
```

# 8. Get the average fare by survival status

```
print(data.groupby('Survived')['Fare'].mean())
```

# 9. Get the number of survivors by class

```
print(data[data['Survived'] == 1]['Pclass'].value_counts())
```

# 10. Get the number of non-survivors by class

```
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

Explorer

Debugger

Average time  
**0.103 s**  
103.00 ms

Maximum time  
**0.103 s**  
103.00 ms

1 out of 1 shown test case(s) passed

Test case 1 103 ms Debug

Expected output

Actual output

|                                |                                |
|--------------------------------|--------------------------------|
| Pclass                         | Pclass                         |
| 1...0.629630                   | 1...0.629630                   |
| 2...0.472826                   | 2...0.472826                   |
| 3...0.242363                   | 3...0.242363                   |
| Name: Survived, dtype: float64 | Name: Survived, dtype: float64 |
| Embarked_S                     | Embarked_S                     |
| 0...0.506073                   | 0...0.506073                   |
| 1...0.336957                   | 1...0.336957                   |
| Name: Survived, dtype: float64 | Name: Survived, dtype: float64 |
| FamilySize                     | FamilySize                     |
| 0...0.303538                   | 0...0.303538                   |
| 1...0.552795                   | 1...0.552795                   |
| 2...0.578431                   | 2...0.578431                   |
| 3...0.724138                   | 3...0.724138                   |
| 4...0.200000                   | 4...0.200000                   |
| 5...0.136364                   | 5...0.136364                   |
| 6...0.333333                   | 6...0.333333                   |
| 7...0.000000                   | 7...0.000000                   |
| 10...0.000000                  | 10...0.000000                  |
| Name: Survived, dtype: float64 | Name: Survived, dtype: float64 |
| IsAlone                        | IsAlone                        |
| 0...0.505650                   | 0...0.505650                   |
| 1...0.303538                   | 1...0.303538                   |
| Name: Survived, dtype: float64 | Name: Survived, dtype: float64 |
| Pclass                         | Pclass                         |
| 1...84.154687                  | 1...84.154687                  |
| 2...20.662183                  | 2...20.662183                  |
| 3...13.675550                  | 3...13.675550                  |
| Name: Fare, dtype: float64     | Name: Fare, dtype: float64     |
| Pclass                         | Pclass                         |
| 1...38.233441                  | 1...38.233441                  |
| 2...29.877630                  | 2...29.877630                  |

Average time  
**0.103 s**  
103.00 ms



Maximum time  
**0.103 s**  
103.00 ms



1 out of 1 shown test case(s) passed



|                                |                                |
|--------------------------------|--------------------------------|
| 5....0.136364                  | 5....0.136364                  |
| 6....0.333333                  | 6....0.333333                  |
| 7....0.000000                  | 7....0.000000                  |
| 10....0.000000                 | 10....0.000000                 |
| Name: Survived, dtype: float64 | Name: Survived, dtype: float64 |
| IsAlone                        | IsAlone                        |
| 0....0.505650                  | 0....0.505650                  |
| 1....0.303538                  | 1....0.303538                  |
| Name: Survived, dtype: float64 | Name: Survived, dtype: float64 |
| Pclass                         | Pclass                         |
| 1....84.154687                 | 1....84.154687                 |
| 2....20.662183                 | 2....20.662183                 |
| 3....13.675550                 | 3....13.675550                 |
| Name: Fare, dtype: float64     | Name: Fare, dtype: float64     |
| Pclass                         | Pclass                         |
| 1....38.233441                 | 1....38.233441                 |
| 2....29.877630                 | 2....29.877630                 |
| 3....25.140620                 | 3....25.140620                 |
| Name: Age, dtype: float64      | Name: Age, dtype: float64      |
| Survived                       | Survived                       |
| 0....30.626179                 | 0....30.626179                 |
| 1....28.343690                 | 1....28.343690                 |
| Name: Age, dtype: float64      | Name: Age, dtype: float64      |
| Survived                       | Survived                       |
| 0....22.117887                 | 0....22.117887                 |
| 1....48.395408                 | 1....48.395408                 |
| Name: Fare, dtype: float64     | Name: Fare, dtype: float64     |
| 1....136                       | 1....136                       |
| 3....119                       | 3....119                       |
| 2....87                        | 2....87                        |
| Name: Pclass, dtype: int64     | Name: Pclass, dtype: int64     |
| 3....372                       | 3....372                       |
| 2....97                        | 2....97                        |
| 1....80                        | 1....80                        |